



Optimization and characterization of toroidal foci for super-resolution fluorescence microscopy: tutorial

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Single-molecule localization microscopy (SMLM) has become an essential tool to investigate phenomena at the nanoscale. Among the different SMLM approaches, methods that interrogate the molecular position with an intensity minimum, such as minimal emission fluxes (MINFLUX) or the more recent raster scanning a minimum of light (RASTMIN), stand out for reaching true molecular resolution. To implement these methods, the phase of the excitation beam needs to be modulated to obtain a focus with a central minimum, i.e., a so-called toroidal or doughnut-shaped focus. In this tutorial, we explain the basis and experimental tricks to generate and optimize such beams, particularly in raster-scanning microscopes. © 2023 Optica Publishing Group

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1. INTRODUCTION

Almost 30 years ago, stimulated emission depletion (STED) microscopy was introduced as one of the pioneering methods in super-resolution microscopy [1]. In STED, the diffraction barrier was broken by reducing the effective size of the focal spot by means of stimulated emission. Therefore, the intensity of the STED or depletion beam needed to be modulated to have a region of minimum, ideally zero, intensity, from which fluorescence could be emitted. After a few years of experimentation with other approaches, [2–6], it was established that toroidal foci presenting a central minimum (zero) of intensity were optimal for two-dimensional STED microscopy [4–6]. Initially, toroidal foci, also known as doughnut-shaped foci, were generated by modulating the phase of the beam with a helical pattern from $0-2\pi$ using spatial light modulators [5–7]. Later, the spatial light modulator was replaced by a spiral phase plate, also known as vortex phase plate (VPP) [8,9]. The use of $0-2\pi$ VPPs became standard to generate foci with a central minimum for STED and reversible saturable optical fluorescence transition (RESOLFT) implementations [10–12].

In recent years, a series of single-molecule localization (SML) methods has been developed in which the position of single fluorescence emitters is determined with nanometer precision using

a sequence of exposures to a light field comprising a minimum of intensity. In this case, the doughnut-shaped focus is used to excite the fluorescence of an emitter instead of suppressing it. Minimal emission fluxes (MINFLUX) was the first reported method of this kind [13], using a sequence of four displaced toroidal foci to interrogate the emitter position. This technique offers 1–2 nm localization precision with a moderate photon budget (1000–2000 photons), enabling an extra push beyond the diffraction barrier for SML microscopy reaching routinely the sub-10 nm realm [14]. MINFLUX has been extended to 3D [15], combined with fluorescence lifetime measurements [16] and with DNA point accumulation in nanoscale topography (DNA-PAINT) [17]. This year, a new implementation of such a method called raster scanning a minimum of light (RASTMIN) was presented [18]. RASTMIN achieves the same nanometer localization precision as MINFLUX but using a raster-scanning microscope (i.e., confocal or two-photon) with two minor modifications. One of the modifications is, of course, the modulation of the excitation beam to obtain a toroidal focus. The other is a drift correction system, necessary for any method aiming to achieve nanometer localization precision/resolution. Remarkably, MINFLUX, RASTMIN, and other SML methods, some recent such as MINSTED [19] and some developed more than 20 years ago such as orbital tracking [20], can be

rationalized under a unique mathematical framework [21]. This common framework for SML through sequentially structured illumination (SML-SSI) enabled a fair comparison and benchmarking of different methods, which demonstrated that methods using light patterns comprising an intensity minimum are significantly more photon efficient [21,22].

The performance of all these methods for SML and super-resolution microscopy relies on the quality of the central intensity minimum, ideally zero, of the beams. Generating such beams requires specific technical knowledge. In this tutorial, we provide a practical step-by-step guide to generate a toroidal focus in a confocal microscope or any raster scanning microscope, and to optimize its quality. We discuss the influence of key experimental parameters that should be finely tuned to obtain an optimum doughnut-shaped focus.

2. HOW TO GENERATE AND MEASURE DOUGHNUT-SHAPED BEAMS

There are several ways of obtaining a toroidal focus. For SML and super-resolution microscopy, it is often preferred to achieve a fluorescence excitation as uniform and isotropic as possible within the focus. For this reason, a focus with circular polarization is usually desired, and methods based on radial polarization to obtain a toroidal focus [23] are not used. By contrast, using a helicoidal $0-2\pi$ phase modulation enables the generation of a toroidal focus with circular polarization. This phase modulation may be introduced in the beam using dynamic devices, such as spatial light modulators or micro-mirror arrays, or helicoidal phase plates, also known as VPPs. The former has the advantage of also enabling the correction of aberrations or switching between phase modulation schemes. The latter provides a fixed configuration but has the advantages of being simpler to implement (it does not require any electronics, software, or computer control), having higher efficiency (typically $>95\%$ in comparison to $70\%-90\%$), and being significantly less costly (1 k USD against 10–20 k USD). Here, we will deal with the use of VPPs to generate and characterize toroidal foci in a laser-scanning microscope such as a confocal microscope. We will first describe the main optical elements to generate a doughnut-shaped focus. Then, we present a simple protocol to characterize experimentally the intensity distribution of the focus and show example data obtained in our confocal setup. At the end of this section, we show how to build a simple tool to determine the polarization state of excitation, the control of which is crucial to optimize the quality of the toroidal focus.

A. Optical Setup to Generate a Toroidal Focus

We show in Fig. 1(a) a scheme of the essential elements needed in the excitation path of a laser-scanning microscope to achieve a toroidal focus. Generating a toroidal focus with helicoidal phase modulation requires a beam with circular polarization in the direction of the helicoidal phase [24,25]. Therefore, along with the VPP, polarization optics should be introduced in the beam path in a place where the beam is collimated.

In the experiments shown in this paper, we used a laser at 642 nm. The polarization state of the beam was adjusted using a broadband quarter-wave plate ($\lambda/4$, Thorlabs AQWP05M)

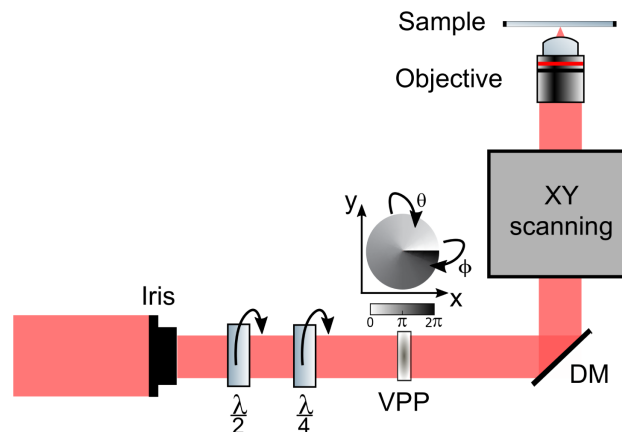


Fig. 1. Scheme of the excitation path with the essential optical elements needed to obtain a toroidal focus in the sample plane. An iris is placed to adjust the beam size. Before the dichroic mirror (DM) that separates the excitation from the fluorescence light, half-wave ($\lambda/2$) and quarter-wave ($\lambda/4$) plates are placed in rotating mounts to adjust the beam polarization. A vortex phase plate (VPP) is mounted on a positioner (x, y) with additional tilt control (φ, θ) to have precise control of the plate position and alignment.

and a broadband half-wave plate ($\lambda/2$, Thorlabs AHWP05M). Both wave plates are mounted in high precision rotation mounts that allow to adjust their axes with respect to the incident beam. It is worth mentioning that the two wave plates are necessary regardless of the polarization state of the incident beam because the polarization state will also be affected by the subsequent elements in the beam path. Dielectric mirrors, although they offer a higher reflectance than metallic mirrors, alter the polarization state of mixed states (not pure s or pure p states) [26]. Silver mirrors instead preserve the state of polarization upon reflection when they are used at 45° . At any other angle, mixed polarization states (such as circularly polarized light) are not preserved. With the $\lambda/2$ and $\lambda/4$, these polarization changes can be compensated for to obtain the desired polarization state, in this case, circular polarization.

The dichroic mirror (DM) used to separate the excitation beam from the emission should be as flat as possible to avoid beam distortion by rugosity and astigmatism. Therefore, we strongly recommend the use of thick DMs (3 mm or 5 mm) since they offer a higher surface flatness [≤ 0.25 waves/inch peak–valley (P-V)] and are less prone to suffer from mechanical stress [27]. We also suggest gluing the DMs on their mounts instead of applying pressure unevenly using screws.

We used a high numerical aperture ($NA = 1.4$) oil immersion objective. The high NA is key to achieving a high collection efficiency of fluorescence photons, which is particularly relevant in single-molecule measurements. However, for excitation, the effective NA may be reduced using an iris. Reducing the illumination NA increases the overall size of the toroidal focus, and in some cases, it facilitates obtaining a high quality central zero of intensity. The reason for this is the fact that aberrations at the edges of a high NA objective are usually more pronounced. Therefore, slight variations in the illumination translate into stronger non-compensated for fields at the central region of the focus.

While there are broadband $\lambda/2$ and $\lambda/4$ available, VPPs are up to now produced for specific wavelengths. We used a VPP designed for 633 nm (V-633-10-1, Vortex Photonics, Germany) made of a fused-silica plate with 64 steps from 0 to 2π . The VPP is mounted onto a five-axis lens positioner (Newport LP-05 A). The positioner allows to control the lateral (x , y) alignment of the VPP as well as its angular tilts (θ , φ) with respect to the optical axis.

It is worth noting that the more optical elements added to the path, the more the polarization and the shape of the beam will be altered. The order of the optical elements (phase plate, polarization optics) described here is exchangeable if they are placed in a portion of the path where the beam is collimated.

B. Characterization of the Focus

The intensity distribution of the focus is determined by scanning a sub-diffraction fluorescence emitter or scatterer over the focus [28,29]. In this way, the small emitter/scatterer probes locally the intensity of the focus, and images over different planes or a tomography of the intensity distribution of the focus can be obtained.

The measured intensity distribution will be the result of the convolution between the actual intensity distribution of the focus and the probe size. From simulations, it can be seen that if the probe is three times smaller than the features of the focus (width of the maxima or minima), the errors in the value of the local intensity are of about 5%. Therefore, we consider suitable fluorescent or scattering probes the ones with a size at least three times smaller than the diffraction limit.

Another requirement for the probes is that they should provide a linear and photostable response. Polystyrene beads containing fluorescent molecules are suitable probes. We used such nanoparticles with a nominal diameter of 40 nm (FluoSpheres Carboxylate-Modified Microspheres, 0,04 μm , Dark Red, Thermo Fisher Scientific). While it is possible to use smaller fluorescent beads of the same kind, they were not suited for our measurement because they underwent significant photobleaching during acquisition.

Since we want a sample to probe the point spread function (PSF) in the same conditions as the SML-SSI experiment, it is necessary to prepare a bead sample surrounded by aqueous media. In this way, we are also reducing the mismatch between the refraction indices of the objective immersion media and the sample. To prepare the sample, we used #1.5 thickness glass-bottomed chamber slides (Lab-Tek II, Thermo Fisher Scientific) and followed these steps:

- (1) treat with 1 M KOH for 10 min to clean the surface (use enough volume to cover the whole glass surface);
- (2) wash three times with Milli-Q water;
- (3) incubate for 15 min with a polydiallyldimethylammonium chloride (PDPA, Sigma-Aldrich) solution (1 mg/mL in 0.5 M NaCl);
- (4) wash three times with Milli-Q water;
- (5) incubate with pre-sonicated bead solution for 15 min;
- (6) wash three times with Milli-Q water.

We included the PDPA incubation step to positively charge the coverslip surface so that the negatively charged beads that we

used would be electrostatically adsorbed. This step depends on the superficial charge of the beads used and can be adapted using a different polymer to coat the surface.

The pre-sonication mentioned in step 5 is advised before each use, to avoid bead aggregation in the sample. The bead solution concentration was adjusted to deliver a density in the sample of approximately 20 beads in a $100 \times 100 \mu\text{m}^2$ field of view. This was achieved by diluting the stock solution in Milli-Q water (1:200,000 dilution). The bead solution was sonicated for 10 min just before being used.

The measurement started by acquiring images of a large field of view (e.g., $100 \times 100 \mu\text{m}^2$) to verify that the bead density was appropriate and that the fluorescence intensity among different beads was homogeneous. In this way, we confirmed that we could address single beads and that the presence of bead aggregates was negligible. For reliable measurements of the focus intensity distribution, exceptionally bright or dim emitters should be avoided since they could represent bead aggregates or other species of unknown size and response. We suggest using fresh samples because this type of bead tends to degrade with time. The acquisition conditions will depend on the setup and the beads used to prepare the sample, but as a rule of thumb, choose the laser power and pixel dwell time to have a signal-to-background ratio (SBR) level >10 and to avoid significant photobleaching during the scanning of a bead.

We show in Fig. 2 images of a bead sample acquired with our setup. Figure 2(a) shows a $100 \times 100 \mu\text{m}^2$ field of view of the sample scanned with the doughnut-shaped focus. Figure 2(b) shows a smaller scan ($10 \times 10 \mu\text{m}^2$) of the region marked in Fig. 2(a) (white square), and Fig. 2(c) the scan of a single bead. A measurement with a higher signal-to-noise ratio (SNR) can be obtained after averaging a series of images. In Fig. 2(d), we show an average of 16 images of the same fluorescent bead.

To evaluate the quality of the toroidal foci obtained in different experimental conditions, we acquire images like the one shown in Fig. 2(d). All the PSFs presented in this tutorial were acquired using a laser power of around 3–5 μW measured at the back focal plane of the objective lens. The dwell time used was 0.5 ms.

C. Evaluation of Incident Beam Polarization

The state of polarization of the incident beam plays an important role in determining the focused field distribution, especially when using high NA objectives. When a high NA objective lens is used, the electric field in the focal region suffers from the effect of depolarization [30,31] and can exhibit a component along the optical axis. Also, to generate a doughnut-shaped focal field using $0-2\pi$ modulation, the polarization of the incident beam must be circular and along the same direction of the phase ramp [24,25].

To facilitate the evaluation of the polarization of the excitation beam, we use a custom-made motorized polarization analyzer [Fig. 3(a)]. It basically consists of a rotating polarizer (Glan Thomson prism coupled to a rotating engine) and a photodiode detector to measure the transmitted intensity. An iris can also be included before the polarizer to adjust the beam size. The voltage signal at the photodiode output can be read with an

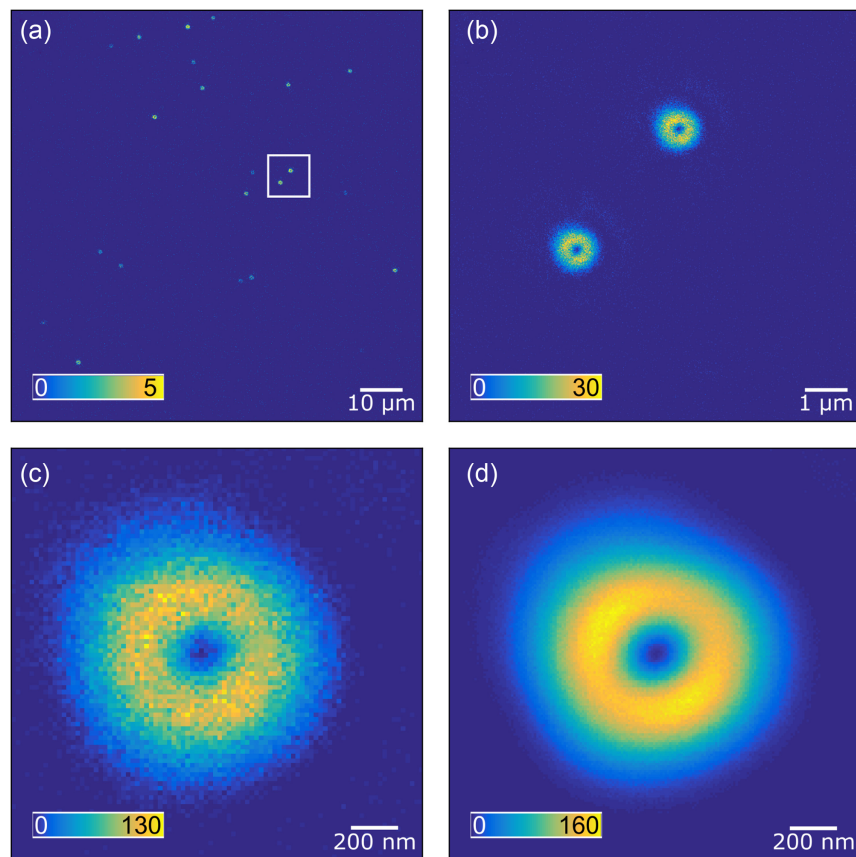


Fig. 2. Measurement of the focus intensity distribution using fluorescent beads. (a) $100 \times 100 \mu\text{m}^2$ field of view, pixel size 67 nm. (b) $10 \times 10 \mu\text{m}^2$ of the area marked with a white square in (a), pixel size 20 nm. (c) Scan over a single bead, pixel size 10 nm. (d) Average intensity (photon counts) of 16 images as the one shown in (c). Color bar units are photon counts.

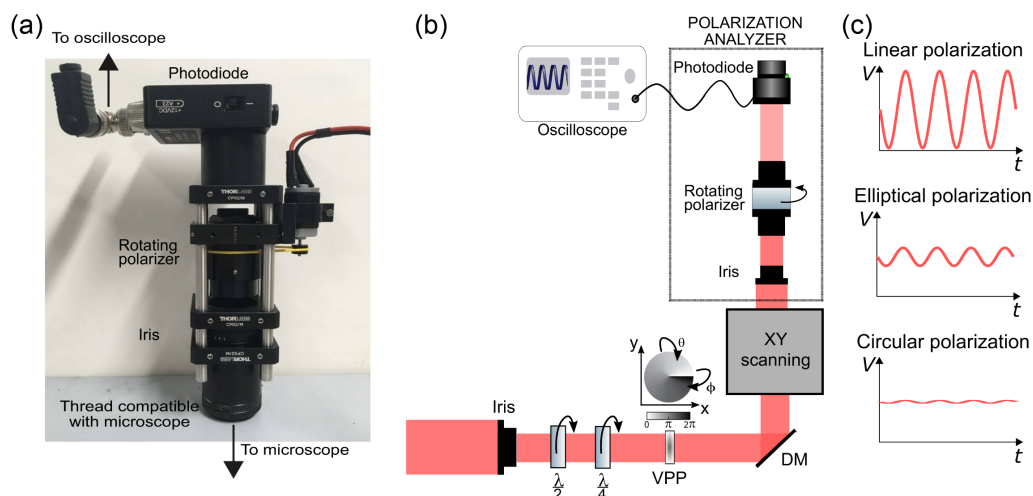


Fig. 3. Polarization analyzer implemented to measure the polarization state of the excitation beam. (a) Picture of the polarization analyzer. (b) Scheme of the polarization analyzer coupled to the microscope and connected to an oscilloscope to measure the output voltage. (c) Example output voltages for different polarization states: signal contrast is maximal for linear polarization and minimal for circular polarization.

oscilloscope or any other voltage acquisition device (Arduino, DAQ).

We used the polarization analyzer to evaluate the polarization state of light just before the objective, as shown in Fig. 3(b). To evaluate the polarization state of the incident beam at the

back focal plane of the objective, the sample holder and the objective were removed, and the objective was replaced with the polarization analyzer.

If the incident beam is linearly polarized, the registered voltage will be modulated at twice the rotation frequency of

the analyzer with maximum amplitude. On the contrary, if the incident beam is circularly polarized (either right-handed or left-handed), the modulation amplitude will be minimum (ideally zero). Anything in between corresponds to elliptical polarization.

To adjust the polarization, both $\lambda/2$ and $\lambda/4$ are rotated while looking at the output signal of the polarization analyzer in an oscilloscope. Alternatively, the registered signal can be analyzed online to compute the contrast as $\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$. The rotation frequency of the polarization analyzer was set to 20 Hz, which is adequate to monitor in real time the modulated signal as the wave plates are rotated.

We show in Fig. 3(c) sketches of typical output signals of the analyzer for linear, elliptical, and circular polarizations. When optimizing for circular polarization, ideally, the modulation should reach a value of zero. However, in real experiments, the minimum modulation contrast achieved is of about 1%–5%, due to the accumulated efficiencies of polarizers and wave plates.

3. OPTIMIZATION OF KEY PARAMETERS

To obtain the most rotationally symmetric doughnut-shaped focus with a central minimum with a value equal to the background signal (minimum intensity), two aspects are critical. First, the polarization of the beam must be circular and with the same hand as the helicoidal phase ramp. Also, the VPP must be precisely aligned in both position and tilt so that its center coincides with the optical axis.

The final aim is to obtain a doughnut-shaped focus that displays:

- (1) a central minimum of intensity with a value equal to the background of the image;
- (2) a narrow minimum of intensity with rotational symmetry.

A. VPP Alignment

To correctly align the phase plate in the excitation path, we use a two-step protocol. First, a coarse alignment of the VPP should be made by looking at the beam profile just after the phase plate using a camera or a screen. We show in Fig. 4(a) example images of the beam profile acquired with a camera in the beam path of our setup. To acquire these images, we introduced a camera in the setup after the VPP positioner. The first image corresponds to the beam profile before adding the VPP. The following three images in Fig. 4(a) correspond to the beam profile after adding the VPP to the path and during the coarse alignment of x and y positions. When the phase plate is misaligned but its center is not significantly off axis with respect to the optical axis, we see in the screen or the camera image the gaussian beam profile but with an intensity minimum.

Modifying the tilt angles will not result in an appreciable change in the beam profile during the coarse alignment. When the intensity zero is in the vicinity of the beam intensity center, we move to the second step to complete the VPP alignment and tilt adjustment by monitoring the intensity distribution of the focus, as explained in Section 2.B.

We show in Figs. 4(c)–4(g) example images of the focus during the alignment process. In every case, only one of the

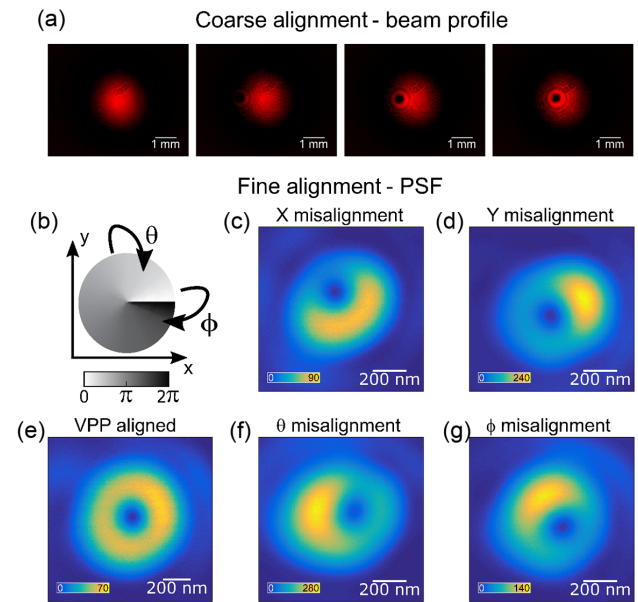


Fig. 4. Coarse and fine alignment of the VPP position and tilt. (a) Images of the free beam profile before introducing the VPP and during coarse alignment. (b) Scheme of the VPP and its four different alignment axes. (c)–(g) Intensity distributions of the focus for different misalignments.

parameters (x , y , θ , ϕ) of the positioner was slightly modified with respect to the optimal VPP alignment [Fig. 4(e)]. Each image is the average of 16 frames of a single bead acquired with a pixel size of 5 nm and scanning a field of view of $1.2 \times 1.2 \mu\text{m}^2$.

B. Polarization

To obtain a toroidal focus with a rotationally symmetric zero intensity, the incident beam should have circular polarization with the same hand as the VPP. We show in Fig. 5 simulations and experimental measurements of the focus intensity distribution for different polarization states of the incident beam: linear, elliptical, and left-handed circular and right-handed circular polarizations. Each experimental image is the average of 16 frames of a single bead acquired with a pixel size of 5 nm and scanning a field of view of $1.2 \times 1.2 \mu\text{m}^2$. The simulated images were calculated with PyFocus [25], a Python package to perform vectorial simulations of focused fields at high NA. Along with the images, we show intensity profiles to better evaluate the quality of the intensity minimum and the symmetry of the focus.

In Fig. 5(a), for right-handed circularly polarized light, the focus has the desired toroidal shape and rotational symmetry. The zero intensity is also rotationally symmetric, and its value corresponds to the intensity of the experimental background. When the state of polarization is changed to left-handed circular, the toroidal shape of the focus vanishes, as can be seen in Fig. 5(b). Only when focusing a beam with right-circular polarization can we obtain the toroidal focus with the desired intensity minimum. This can be explained by looking at the different vectorial components of the focused field. Although the lateral components of the electric field have a minimum in the focal region for any polarization, the axial component has a

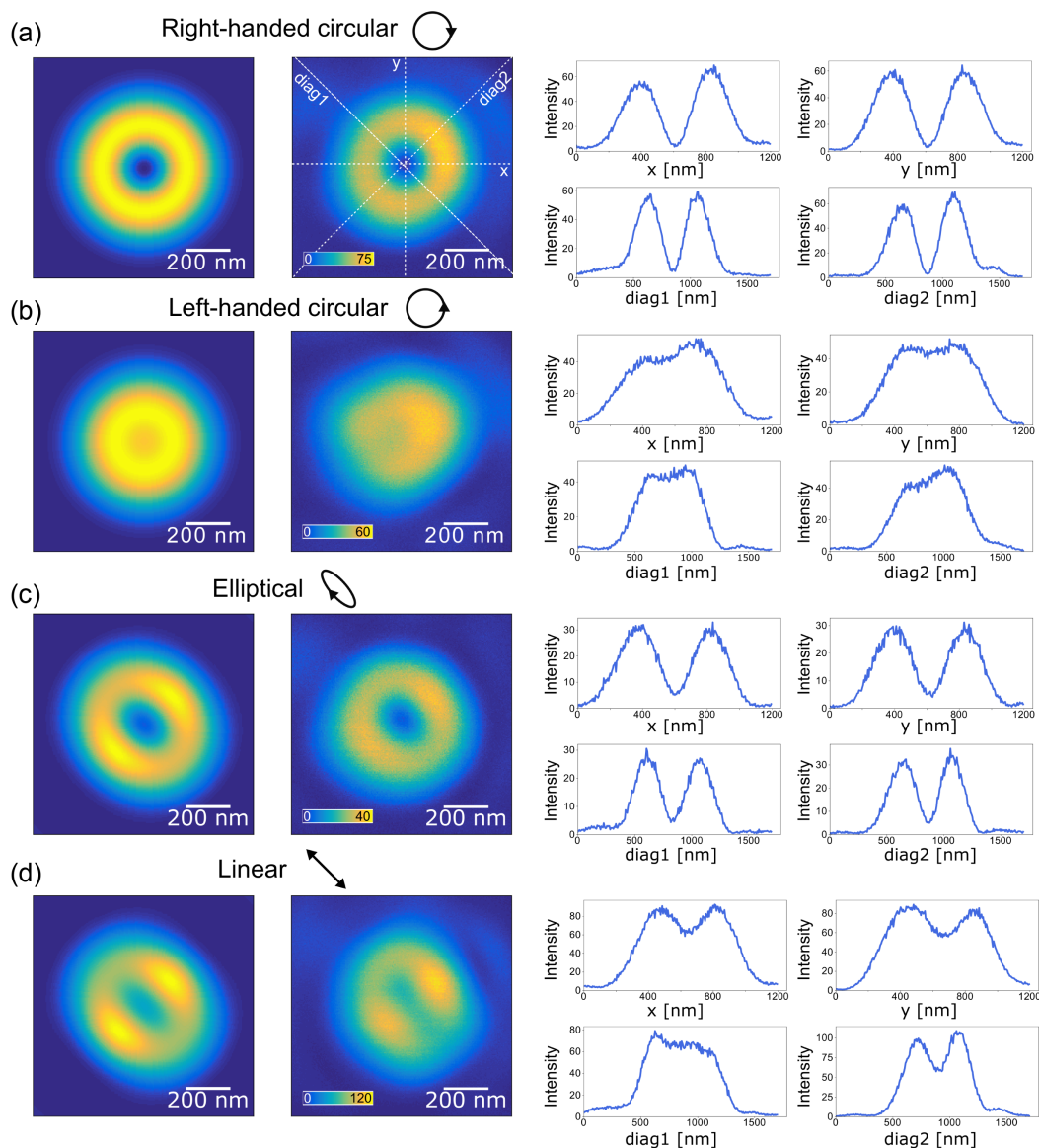


Fig. 5. Simulated (first column) and experimental (second column) focus intensity distributions for the different polarization states of light and corresponding profiles of the experimental intensity distributions along the four axes indicated in (a) (white dotted lines). (a) Right-handed circular polarization. (b) Left-handed circular polarization. (c) Elliptical polarization. (d) Linear polarization. For the experimental focus, pixel size is 5 nm and dwell time is 0.5 ms. Parameters of the simulated focus field generated with PyFocus: NA = 1.4, $n = 1.5$, objective radius = 3 mm, wavelength = 640 nm, resolution = 5 nm.

minimum only for circular polarization with the same hand as the helicoidal phase. For circular polarization with the opposite hand, the axial component presents a central maximum. For elliptical polarization, depending on the degree of ellipticity, the intensity minimum in the center is maintained but its value deviates from the background value. In foci generated with low NA, since the axial component of the electric field has a small relative amplitude when compared to the lateral components, this effect may not be significant. Experimentally, it is not easy to observe the hand of the VPP. Instead, once the polarization of the beam is adjusted to be circular, the VPP is roughly aligned, and if a central zero is not observed, it should be inverted.

C. Numerical Aperture

Another aspect that shapes the focus and the quality of the intensity minimum is the size of the beam (effective NA). To control this parameter, we introduce an iris in the path as shown in the setup scheme of Fig. 1. If the full NA is used and the objective pupil is filled or overfilled, then the tightest focus is produced. By closing the iris, the effective focusing NA is reduced and overall size of the focus increases.

We show this effect in Fig. 6 where images of the focus intensity were acquired using the full objective NA [Fig. 6(a)] and using a reduced beam size [low NA, Fig. 6(b)]. In addition to the images, intensity profiles along the diagonals are shown. By comparing both the focus images and the intensity profiles, it is evident that, in addition to increasing the size of the focus,

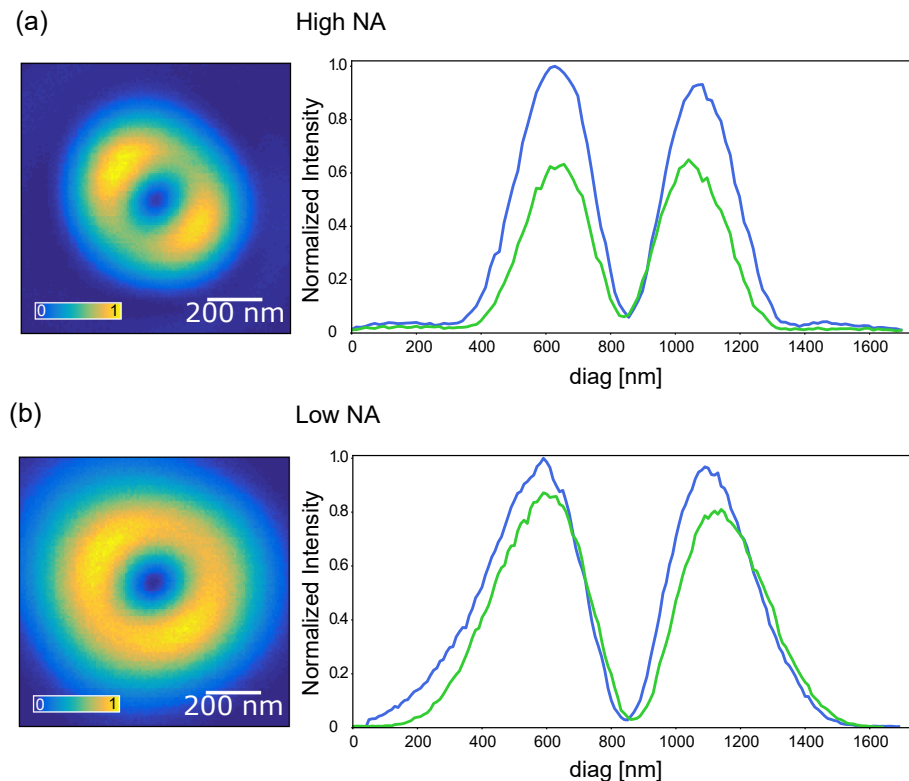


Fig. 6. Effect of decreasing the effective NA of the excitation beam on the doughnut focus. (a) Normalized intensity distribution of the focus and profiles along the diagonals indicated in Fig. 5(a) for the full NA case ($NA = 1.4$). (b) Normalized intensity distribution of the focus and profiles along the diagonals when the beam size is reduced, and we use a lower effective $NA = 1.2$.

lowering the effective NA of the beam also improves the angular symmetry of the toroidal focus. It is interesting to have this as a parameter that can be tuned according to the application of the toroidal focus. If wanting a tighter focus with a narrower zero, one should use the full NA, but if this is not a requirement and instead the doughnut symmetry is more important, then the beam size can be reduced.

4. CONCLUSIONS

We presented a tutorial guide, with practical tips and tricks to generate toroidal foci in a raster-scanning microscope by including a VPP and having precise control of the state of polarization of the excitation beam. We show a simple way to adjust the state of polarization of the excitation beam that is also useful in any microscope. Also, we provide detailed protocols to characterize experimentally the intensity distribution of the focus and to adjust the most critical parameters to have good quality toroidal foci.

While we restricted our guide to working with VPPs, some of the more general aspects of the beam generation and characterization are also useful when working with other phase modulators such as spatial light modulators or micro-mirror arrays.

We believe that this tutorial is a useful tool for microscopists who want to expand the capacities of their setups, particularly to perform STED, MINIFLUX, RASTMIN, or other methods involving foci comprising a minimum of intensity. In this

way, we hope to contribute to expanding the access to these super-resolution methods.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

1. S. W. Hell and J. Wichmann, *Breaking the Diffraction Resolution Limit by Stimulated Emission: Stimulated-Emission-Depletion Fluorescence Microscopy* (1994), Vol. **19**.
2. T. A. Klar and S. W. Hell, *Subdiffraction Resolution in Far-Field Fluorescence Microscopy* (1999), Vol. **24**.
3. T. A. Klar, S. Jakobs, M. Dyba, A. Egner, and S. W. Hell, "Fluorescence microscopy with diffraction resolution barrier broken by stimulated emission," *Proc. Natl. Acad. Sci. USA* **97**, 8206–8210 (2000).
4. P. Török and P. R. T. Munro, "The use of Gauss-Laguerre vector beams in STED microscopy," *Opt. Express* **12**, 3605 (2004).
5. K. I. Willig, S. O. Rizzoli, V. Westphal, R. Jahn, and S. W. Hell, "STED microscopy reveals that synaptotagmin remains clustered after synaptic vesicle exocytosis," *Nature* **440**, 935–939 (2006).
6. K. I. Willig, J. Keller, M. Bossi, and S. W. Hell, "STED microscopy resolves nanoparticle assemblies," *New J. Phys.* **8** (2006).
7. K. I. Willig, B. Harke, R. Medda, and S. W. Hell, "STED microscopy with continuous wave beams," *Nat. Methods* **4**, 915–918 (2007).

8. T. Watanabe, M. Fujii, Y. Watanabe, N. Toyama, and Y. Iketaki, "Generation of a doughnut-shaped beam using a spiral phase plate," *Rev. Sci. Instrum.* **75**, 5131–5135 (2004).
9. M. W. Beijersbergen, R. P. C. Coerwinkel, M. Kristensen, and P. Woerdman, *Helical-Wavefront Laser Beams Produced with a Spiral Phaseplate* (1994), Vol. **112**.
10. G. Vicidomini, P. Bianchini, and A. Diaspro, "STED super-resolved microscopy," *Nat. Methods* **15**, 173–182 (2018).
11. P. Bianchini, C. Peres, M. Oneto, S. Galiani, G. Vicidomini, and A. Diaspro, "STED nanoscopy: a glimpse into the future," *Cell Tissue Res.* **360**, 143–150 (2015).
12. T. Müller, C. Schumann, and A. Kraegeloh, "STED microscopy and its applications: new insights into cellular processes on the nanoscale," *ChemPhysChem* **13**, 1986–2000 (2012).
13. F. Balzarotti, Y. Eilers, K. C. Gwosch, A. H. Gynnå, V. Westphal, F. D. Stefani, J. Elf, and S. W. Hell, "Nanometer resolution imaging and tracking of fluorescent molecules with minimal photon fluxes," *Science* **355**, 606–612 (2017).
14. L. A. Masullo, A. M. Szalai, L. F. Lopez, and F. D. Stefani, "Fluorescence nanoscopy at the sub-10 nm scale," *Biophys. Rev.* **13**, 1101–1112 (2021).
15. K. C. Gwosch, J. K. Pape, F. Balzarotti, P. Hoess, J. Ellenberg, J. Ries, and S. W. Hell, "MINFLUX nanoscopy delivers 3D multicolor nanometer resolution in cells," *Nat. Methods* **17**, 217–224 (2020).
16. L. A. Masullo, F. Steiner, J. Zähringer, L. F. Lopez, J. Bohlen, L. Richter, F. Cole, P. Tinnefeld, and F. D. Stefani, "Pulsed Interleaved MINFLUX," *Nano Lett.* **21**, 840–846 (2021).
17. L. M. Ostersehl, D. C. Jans, A. Wittek, J. Keller-Findeisen, K. Inamdar, S. J. Sahl, S. W. Hell, and S. Jakobs, "DNA-PAINT MINFLUX nanoscopy," *Nat. Methods* **19**, 1072–1075 (2022).
18. L. A. Masullo, A. M. Szalai, L. F. Lopez, M. Pilo-Pais, G. P. Acuna, and F. D. Stefani, "An alternative to MINFLUX that enables nanometer resolution in a confocal microscope," *Light Sci. Appl.* **11**, 199 (2022).
19. M. Weber, M. Leutenegger, S. Stoldt, S. Jakobs, T. S. Mihaila, A. N. Butkevich, and S. W. Hell, "MINSTED fluorescence localization and nanoscopy," *Nat. Photonics* **15**, 361–366 (2021).
20. J. Enderlein, "Tracking of fluorescent molecules diffusing within membranes," *Appl. Phys. B* **71**, 773–777 (2000).
21. L. A. Masullo, L. F. Lopez, and F. D. Stefani, "A common framework for single-molecule localization using sequential structured illumination," *Biophys. Rep.* **2**, 100036 (2022).
22. P. Zdańkowski, L. F. Lopez, G. P. Acuna, and F. D. Stefani, "Nanometer resolution imaging and tracking of single fluorophores by sequential structured illumination," *ACS Photon.* **9**, 3777–3785 (2022).
23. H. Kang, B. Jia, and M. Gu, "Polarization characterization in the focal volume of high numerical aperture objectives," *Opt. Express* **18**, 10813–10821 (2010).
24. X. Hao, C. Kuang, T. Wang, and X. Liu, "Effects of polarization on the de-excitation dark focal spot in STED microscopy," *J. Opt.* **12**, 115707 (2010).
25. F. Caprile, L. A. Masullo, and F. D. Stefani, "PyFocus—a python package for vectorial calculations of focused optical fields under realistic conditions. Application to toroidal foci," *Comput. Phys. Commun.* **275**, 108315 (2022).
26. M. Delay, *Practical Aspects of Mirror Usage in Optical Systems for Biology* (2020).
27. Chroma Technology Corp., Tech Note Dichroics, <https://www.chroma.com>.
28. P. J. Shaw and D. J. Rawlins, "The point-spread function of a confocal microscope: its measurement and use in deconvolution of 3-D data," *J. Microsc.* **163**, 151–165 (1991).
29. R. W. Cole, T. Jinadasa, and C. M. Brown, "Measuring and interpreting point spread functions to determine confocal microscope resolution and ensure quality control," *Nat. Protoc.* **6**, 1929–1941 (2011).
30. K. Bahlmann and S. W. Hell, "Electric field depolarization in high aperture focusing with emphasis on annular apertures," *J. Microsc.* **200**, 59–67 (2000).
31. D. Ganic, X. Gan, and M. Gu, "Focusing of doughnut laser beams by a high numerical-aperture objective in free space," *Opt. Express* **11**, 2747–2752 (2003).